

Three Motions That Shape Our Lives S6E2

Motion	What Moves	How Long	What It Causes
Earth ROTATES	Earth spins on its axis	1 day (24 hrs)	Day and night
Moon ORBITS Earth	Moon travels around Earth	About 1 month	Phases of the Moon
Earth ORBITS Sun	Earth travels around the Sun (revolution)	1 year (365¼ days)	The seasons

Satellite = any object that orbits a larger object. The Moon is Earth's **natural** satellite; GPS/TV satellites are **artificial**.

Why the Moon Has Phases S6E2a

The Moon makes **no light of its own** — it's a giant mirror reflecting sunlight. The Sun always lights **exactly half** of the Moon. As the Moon orbits Earth, we see that lit half from **different angles** — the shape we see is a **phase**.

✓ The Key Idea

The Moon never changes shape — it's always a round ball. What changes is **how much of the lit half WE can see**. Phases are about our point of view!

✗ Common Mix-Up

Phases are **NOT caused by Earth's shadow!** The dark part is simply the half the Sun isn't lighting at that angle. (Earth's shadow only matters during a **lunar eclipse**.)

The Eight Moon Phases S6E2a



Word	Meaning	What's Happening
Waxing	Getting bigger	Lit part grows from new moon toward full moon
Waning	Getting smaller	Lit part shrinks from full moon back toward new moon
Crescent	A sliver	Less than half of the Moon is lit
Gibbous	More than half	More than half (but not all) of the Moon is lit

Memory trick: Waxing = light "waxing **on**" (growing); Waning = "wane **away**" (shrinking). In the Northern Hemisphere: **right side lit = waxing**, left side lit = waning.

ACTIVITY 1: Phase Order!

Number these phases in order, starting with the new moon (1):

____ Full Moon ____ First Quarter ____ Waning Gibbous
 ____ New Moon ____ Waxing Crescent ____ Waxing Gibbous

ACTIVITY 2: Waxing or Waning?

Write **WAX** (growing) or **WANE** (shrinking):

____ 1. The Moon two days after a new moon ____ 3. The lit part grows bigger each night
 ____ 2. The Moon two days after a full moon ____ 4. Heading from full moon to last quarter

Flip over for eclipses and the **REAL** reason we have seasons! →

Solar & Lunar Eclipses S6E2b

An **eclipse** = the Sun, Earth, and Moon line up in a straight row, so one object's **shadow** falls on another. The **middle object casts the shadow**.

SOLAR ECLIPSE

Order: **Sun** → **MOON** → **Earth**. The Moon is in the middle; its shadow falls on **Earth**. The daytime sky goes dark — the **Sun gets blocked**.

LUNAR ECLIPSE

Order: **Sun** → **EARTH** → **Moon**. Earth is in the middle; its shadow falls on the **Moon**, which often turns dim red — the **Moon gets darkened**.

 **Keep them straight:** name the eclipse after the object that gets **blocked/darkened**. **? Why not every month?** The Moon's orbit is slightly **tilted** — most months the shadow misses, passing a little above or below.

ACTIVITY 3: Solar or Lunar?

Write **S** (solar) or **L** (lunar):

- | | |
|---|--|
| <input type="text"/> 1. The Moon is between the Sun and Earth | <input type="text"/> 4. The daytime sky goes dark for a short time |
| <input type="text"/> 2. Earth's shadow falls on the Moon | <input type="text"/> 5. Order: Sun → Earth → Moon |
| <input type="text"/> 3. The Moon turns a dim reddish color | <input type="text"/> 6. The Moon casts the shadow |

Why We Have Seasons: TILT, Not Distance! S6E2c

Earth is tilted on its axis by about **23.5°** — and the tilt stays pointed the **same direction** all year as Earth orbits the Sun.

When your hemisphere...	Sunlight is...	Days are...	Season
Tilts toward the Sun	Direct and concentrated	Longer	Summer
Tilts away from the Sun	Spread out and weak (slanted)	Shorter	Winter

NOT distance!

Proof: when it's **summer** in the **Northern Hemisphere**, it's **winter** in the **Southern** — at the same time! If distance mattered, the whole planet would share seasons. Earth is actually **closest to the Sun** in **early January** — during northern winter!

Direct vs. slanted light

Direct sunlight (straight down) concentrates energy = strong heating. **Slanted** sunlight spreads the same energy over more ground = weak heating. Same Sun, different angle!

Time of Year	Northern Hemisphere	Southern Hemisphere
June	Summer (tilted toward Sun)	Winter (tilted away)
December	Winter (tilted away)	Summer (tilted toward Sun)

ACTIVITY 4: Season Sense

Circle **T** (true) or **F** (false):

- | | |
|---|--|
| T / F 1. Summer happens when Earth is closest to the Sun | T / F 4. Both hemispheres have summer at the same time |
| T / F 2. Earth's axis is tilted about 23.5° | T / F 5. When it's summer in Georgia, it's winter in Australia |
| T / F 3. Direct sunlight heats more than slanted sunlight | T / F 6. With no tilt, there would be no seasons |

I Can... (Self-Check)

- | | |
|---|--|
| ☐ Match each motion (rotation, Moon's orbit, revolution) to what it causes | ☐ Tell waxing from waning, crescent from gibbous |
| ☐ Explain where the Moon's light really comes from | ☐ Diagram solar and lunar eclipses (who's in the middle!) |
| ☐ Explain why phases are NOT caused by Earth's shadow | ☐ Explain that TILT — not distance — causes seasons |
| ☐ Name the 8 phases in order | ☐ Explain why hemispheres have opposite seasons |